Time-Domain Systems and THz Material Characterization

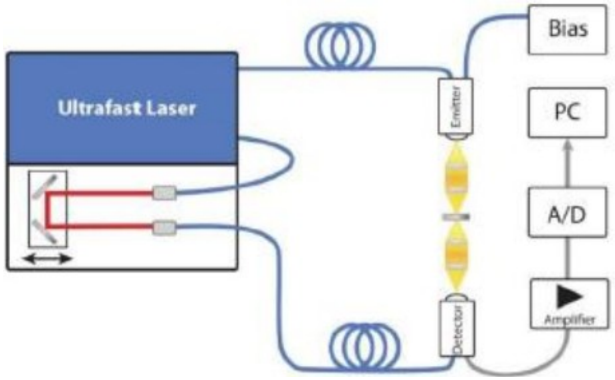
EE4730 Oral Exam | Time-Domain Topic | Daniel Tyukov | July 2026

Why measure in the time domain

- The THz band (0.3 to 3 THz) sits above transistor f_{max} (records around 0.75 THz) and below optical sources. Generating and detecting power is the hard part.
- Two generation paradigms from the lectures: multiply a microwave oscillator up in frequency, or switch a DC bias with a femtosecond laser in a photoconductor.
- The photoconductive route gives a single picosecond pulse whose spectrum covers the entire band at once.
- Measure the field E(t), take an FFT, and one scan characterizes a material from 0.15 to 3 THz. That is what this lab does.

	Heterodyne	Photoconductive
Source	LO chain, xN	fs laser + DC bias
Signal	CW tone	ps pulse
Bandwidth	~30 GHz	up to ~2 THz
Use here	VNA labs	this lab

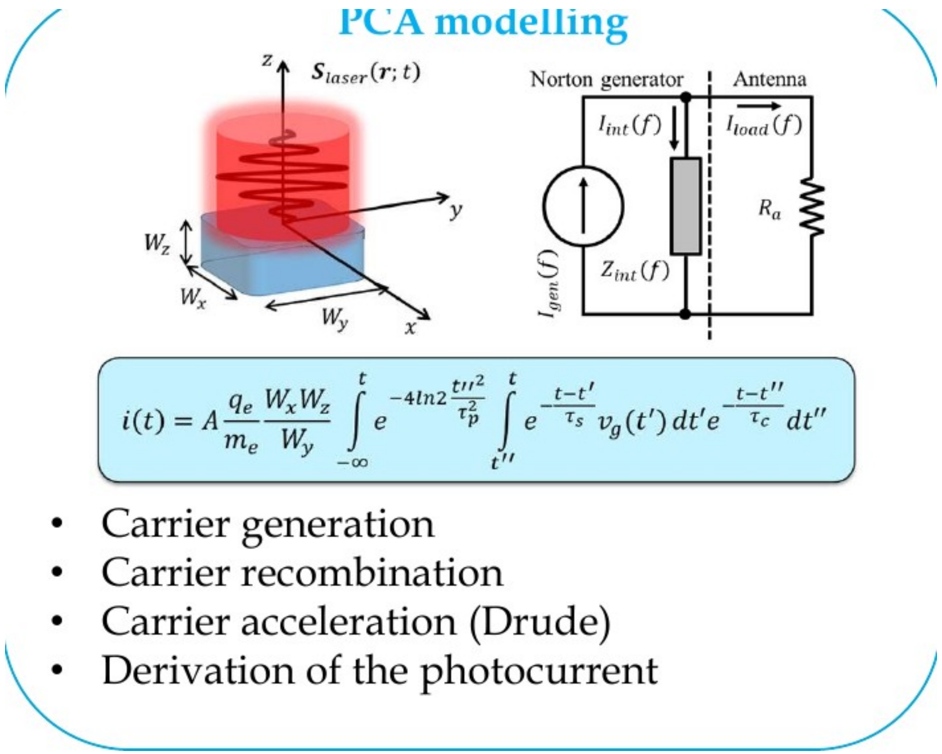
Lecture 2: the two ways to make THz signals



TDS spectrometer chain (lab manual)

The photoconductive antenna

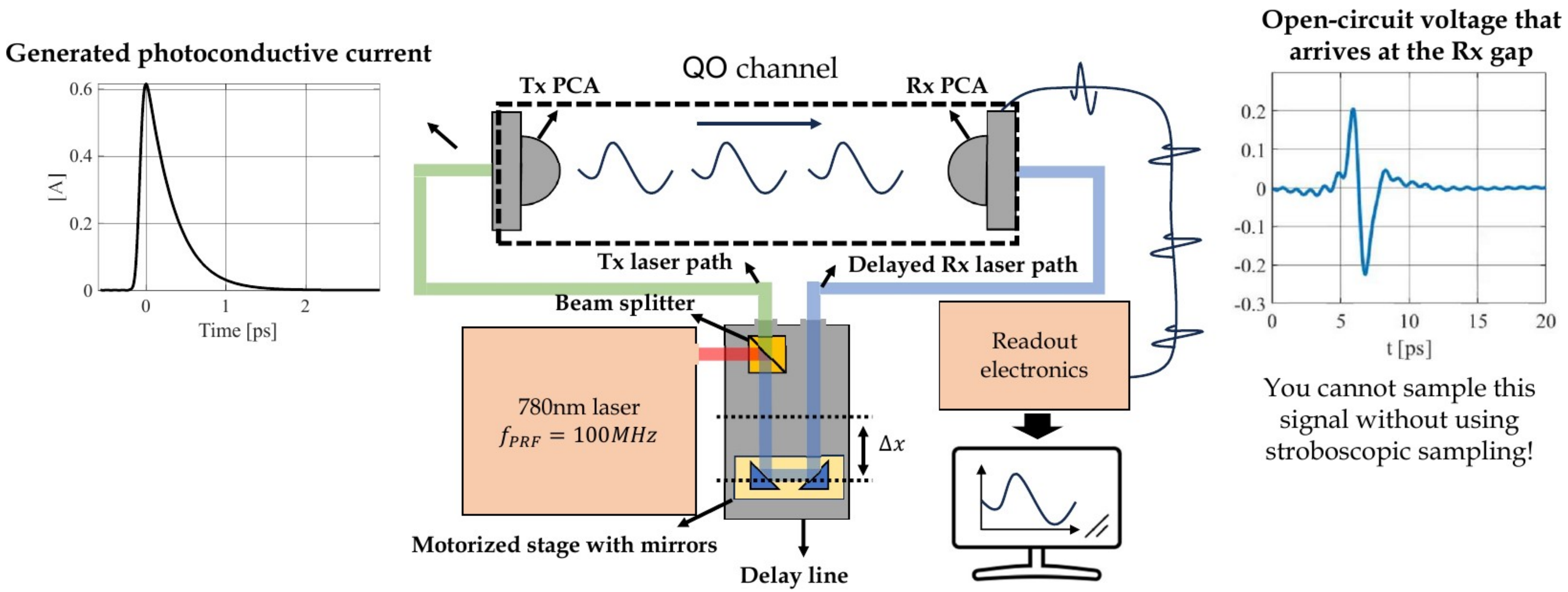
- An LT-GaAs gap between two electrodes. The 1.59 eV band gap matches the 780 nm pump laser.
- The femtosecond pulse creates carriers, the DC bias accelerates them, and recombination cuts the current off after about 1 ps. That current burst radiates the THz pulse.
- The lecture photocurrent model has exactly three ingredients: carrier generation (Gaussian laser pulse), Drude acceleration, and exponential recombination with τ_c.
- The receiver is the same device without bias. The incident THz field accelerates the carriers, so the gated current is proportional to the field at the moment of illumination.
- At high optical power the radiated field screens the bias and the photocurrent saturates. That is why high-power sources use arrays of matched PCAs.



PCA model and Norton equivalent (Lecture 2)

Stroboscopic sampling

- No ADC samples a THz waveform directly; electronics stop at a few gigasamples per second.
- The laser fires an identical pulse every 10 ns (100 MHz repetition). The receiver gap conducts only during the picosecond it is illuminated, so each repetition samples the field at one delay.
- A motorized mirror sweeps the delay (1 mm of travel adds 6.67 ps of round trip). The readout measures a DC average current $i_{rec}(\Delta t)$ proportional to the field at the gate instant; sweeping Δt reconstructs $E(t)$.



Full TDS chain: one laser, a beam splitter, and a delay line (Lecture 2)

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Noise: what the lecture predicts

- Average transmit power 1 mW at 100 MHz repetition: 10 pJ per pulse, roughly 20 W of peak power.
- With link efficiency near 1/10, the receiver current burst is about 0.14 A for half a picosecond.
- The duty cycle dilutes it: $i_{ave} = i_{peak} \cdot \tau_{rec}/T_{rep} \approx 7\ \mu A$ of DC current at the amplifier input.
- The readout amplifier noise is about 70 pA, so a single sweep already delivers a signal-to-noise ratio near 100 dB.
- Averaging N sweeps beats down white noise only:

$$\mathbf{SNR_N = SNR_1 + 10 \cdot \log_{10} N}$$

Prediction for the lab

N = 50 → +17.0 dB

N = 100 → +20.0 dB

N = 1000 → +30.0 dB

Assignment 1 tests exactly this.

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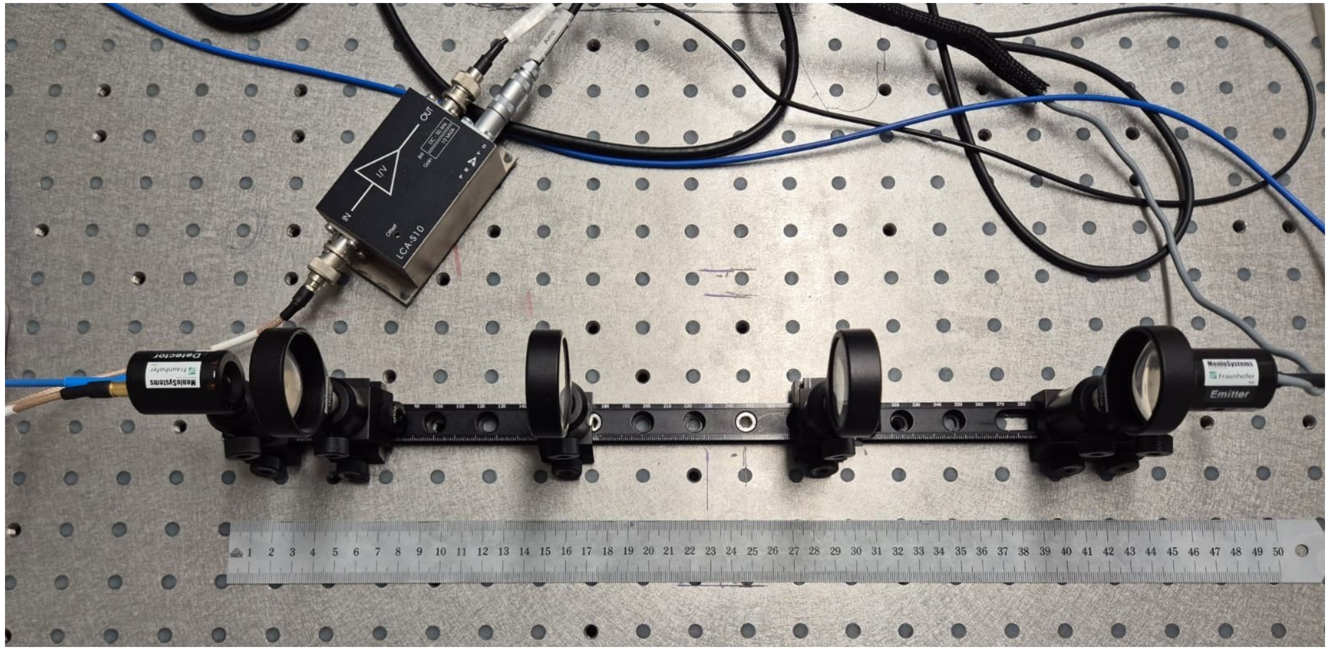
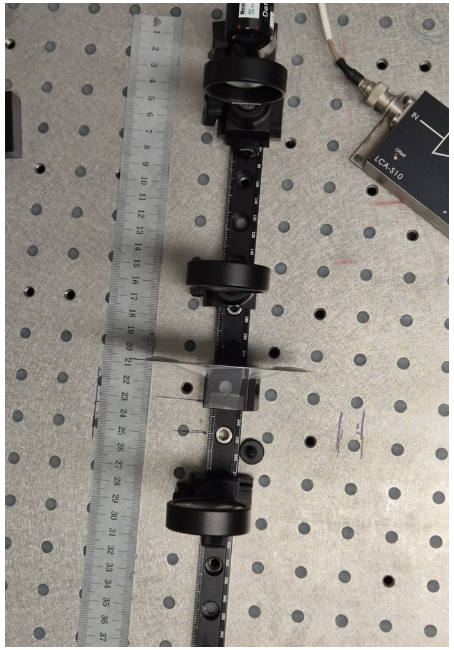
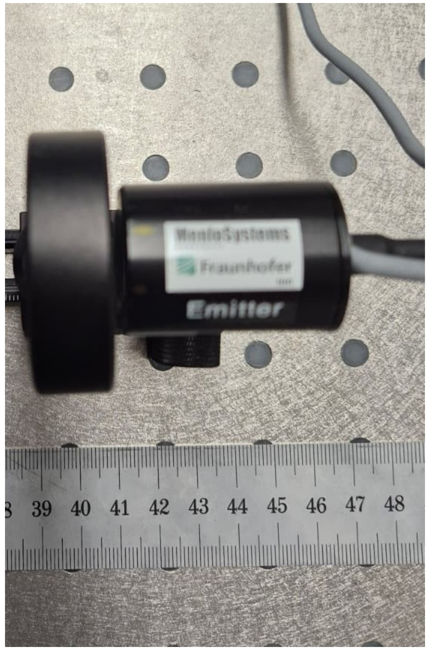
The lab bench



Emitter (right), four TPX50 lenses, detector and LCA-S10 amplifier (left)



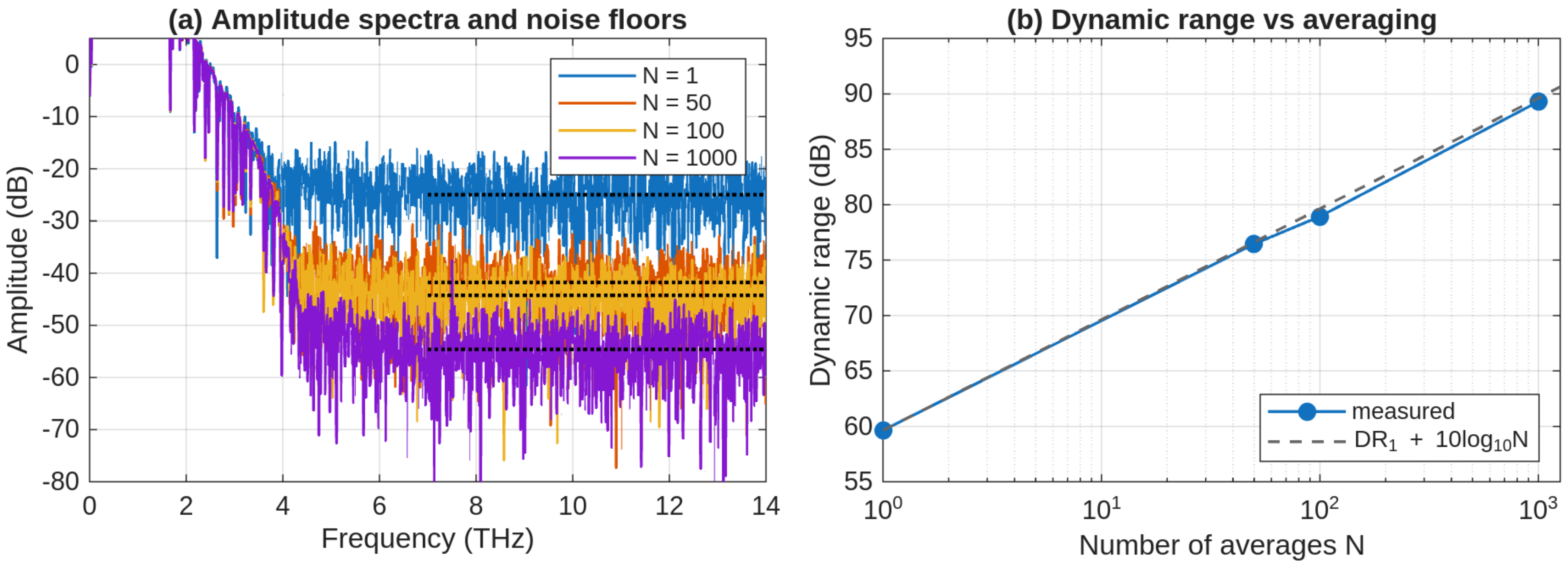
Slab at the pinhole focus



Emitter PCA on its carrier

- Both PCAs are fibre-coupled to the Menlo unit. Lenses 1 and 4 collimate at the antennas, lenses 2 and 3 form an intermediate focus where the samples go (carriers at 35 / 70 / 190 / 245 / 305 / 400 / 435 mm).
- Alignment by maximizing detector current lens by lens; the aligned pulse peaks at 2.89 a.u.
- Every scan: 100 ps delay window, 33.3 fs steps, Tukey window ($\alpha = 0.1$) in ScanControl.

Averaging, measured against the prediction



- Dynamic range grows from 59.6 dB (N = 1) to 89.3 dB (N = 1000): gains of 16.9 / 19.3 / 29.7 dB against the predicted 17 / 20 / 30 dB. The white-noise law holds within 0.7 dB.
- The time trace saturates: past N = 100 the pre-pulse floor stays near $4\cdot10^{-4}$ of the peak because a coherent baseline structure repeats in every sweep. Averaging only buys bandwidth where the noise is white.

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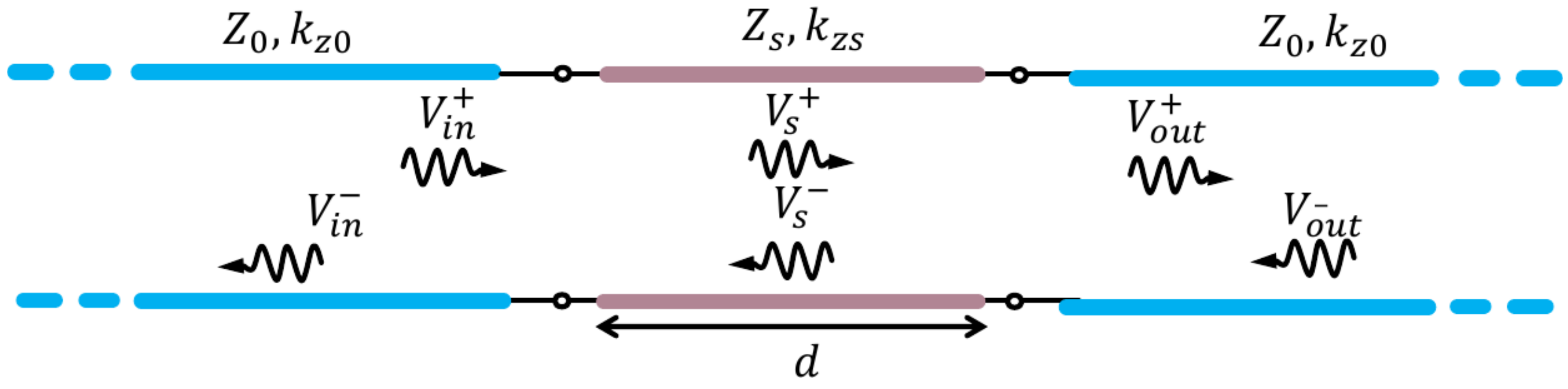
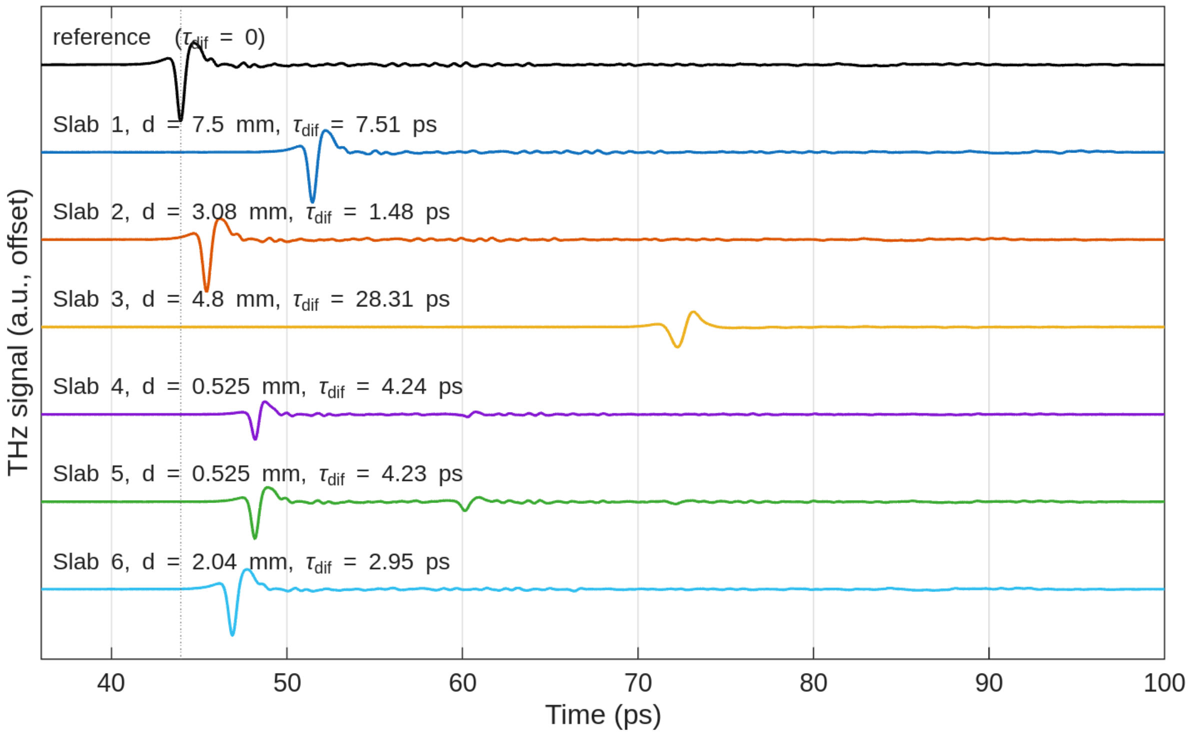
Extracting permittivity and loss

- Differential measurement: a reference scan without the sample, a sample scan with it, and their ratio $H(f) = E_{sam}(f) / E_{ref}(f)$. The $N = 100$ trace is the reference.
- Delay gives the permittivity: the slab replaces a thickness d of air, so the pulse arrives $\Delta\tau = (n-1)\cdot d/c_0$ later and $\varepsilon_r = n^2$.
- Loss comes from the spectrum: model the slab as a transmission-line section between air lines, then fit tan $\delta(f)$ until the modelled magnitude matches the measurement.
- The model contains the Fabry-Perot etalon of the slab faces. If the internal echo lands outside the 100 ps window (slabs 1 and 3), only the first transit is kept.



Air / slab / air transmission-line model from the processing notes

Six slabs identified

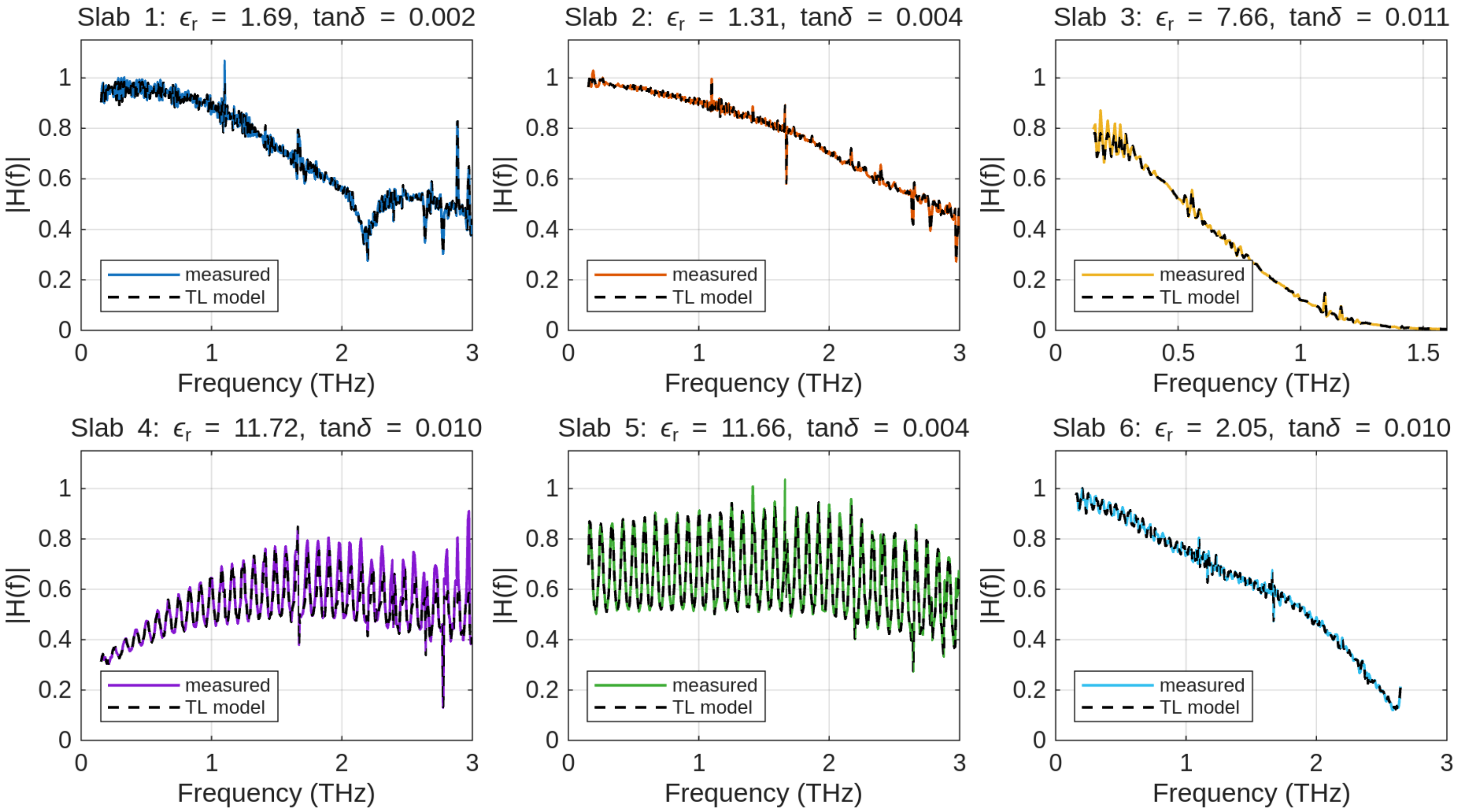


Reference and slab pulses; the delay carries ε_r

Slab	d (mm)	ε_r	tan δ (1 THz)	Material
1	7.5	1.69	0.0010	porous PTFE *
2	3.08	1.31	0.0026	Gore-Tex
3	4.8	7.66	0.0131	Ceramic
4	0.525	11.72	0.0138	Silicon
5	0.525	11.66	0.0028	Silicon
6	2.04	2.05	0.0082	Teflon

- Five of six match the manual's reference table within 3.1%.
- Slab 3 arrives 28 ps late and heavily attenuated; the silicon wafers show their etalon echo 12 ps after the main pulse.
- * Slab 1 (ε_r = 1.69) sits between Gore-Tex and Teflon: consistent with porous PTFE of intermediate density.

Transfer function fits



- The fit reproduces level and etalon ripple; on silicon the ripple period $c_0/(2nd) \approx 83$ GHz pins the optical thickness.

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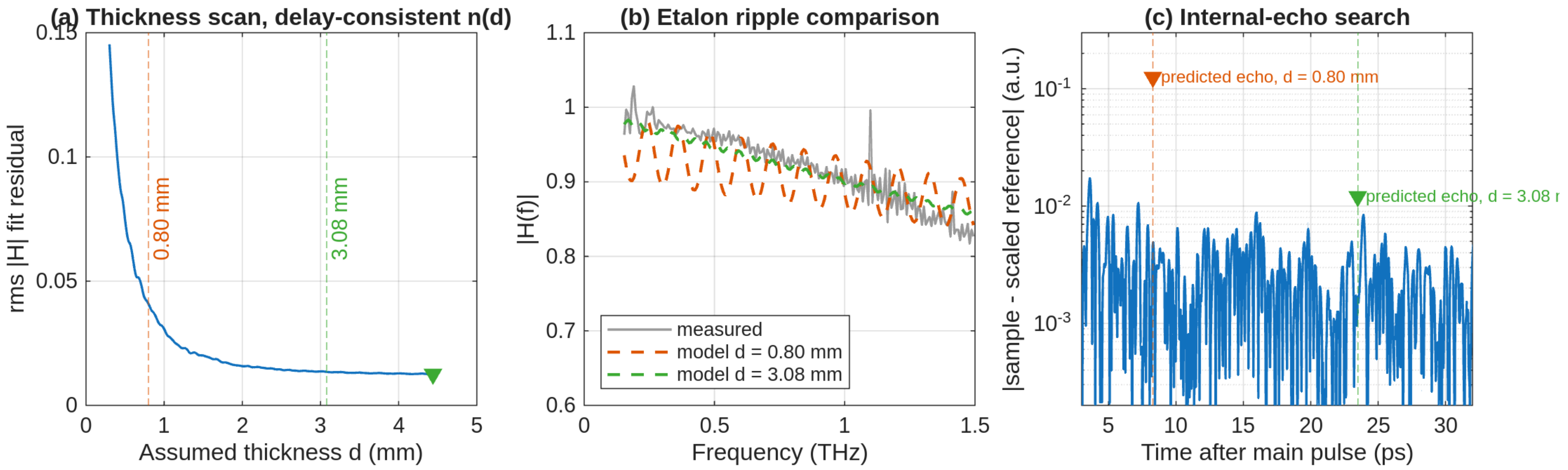
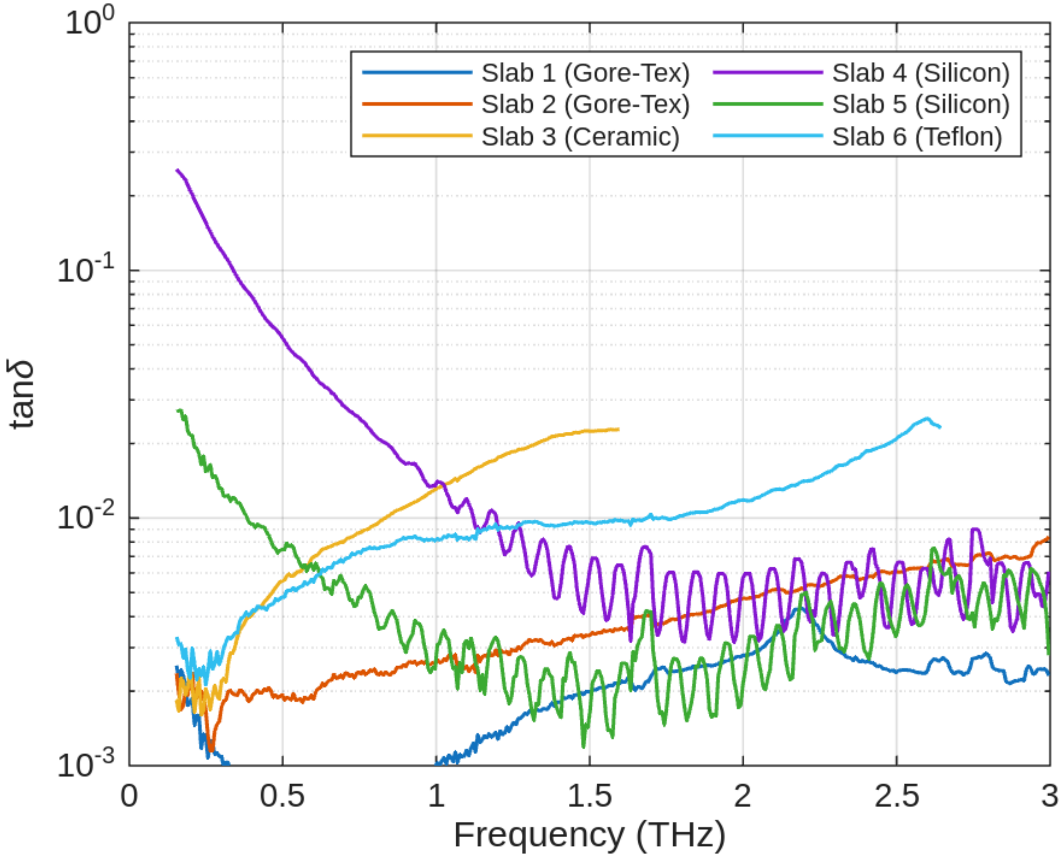
The slab 2 thickness conflict



- The data file says 0.8 mm, the label on the holder says 3.08 mm. The delay alone cannot decide: both give a plausible material (ε_r = 2.44, HDPE-like, versus 1.31, Gore-Tex-like).
- Three independent tests reject 0.8 mm: the fit residual is three times worse, the required ±10% ripple at 120 GHz is absent, and the predicted internal echo at +8.3 ps (0.13 a.u.) does not exist in the residual.
- Conclusion: d = 3.08 mm and the slab is Gore-Tex. The file name recorded a different sample or a typo.

Silicon wafers: same index, different loss



Extracted tan δ(f): silicon falls as 1/f, polymers rise

- Both wafers give n ≈ 3.42, within 0.3% of the literature value for high-resistivity silicon. The etalon ripple period confirms the permittivity independently.
- Their loss differs by an order of magnitude. Conduction loss follows $\tan\delta = \sigma/(\omega\cdot\varepsilon_0\cdot\varepsilon_r)$, so constant conductivity appears as a 1/f slope. That is the measured behaviour below 1.5 THz.
- Converting the median conductivity of wafer 5 to resistivity gives 37 Ω·cm. The wafer is labelled 33 Ω·cm from a four-point-probe measurement: an independent end-to-end check of the loss extraction.
- Wafer 4 comes out at 9 Ω·cm, the more doped and lossier twin.

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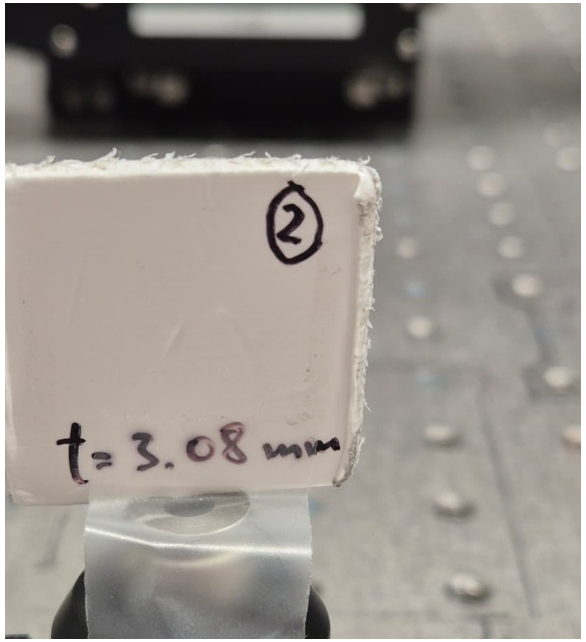
What this lab showed

- Stroboscopic sampling works as taught: a DC photocurrent measurement reconstructs the THz field with 33 fs steps.
- The averaging law holds: 29.7 dB measured against 30 dB predicted at N = 1000. What it cannot remove is coherent structure, and the time-domain floor shows exactly that.
- One delay reading plus a transmission-line fit identified five of six materials within 3.1% of the reference permittivities.
- The same physics settles a metadata conflict (slab 2) and reproduces a wafer's labelled resistivity within 12%.



Slab 2, marked t = 3.08 mm

Thank you

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